

IT Security 2

Unit 13 - Using Decryption to Block Threats in Encrypted Traffic

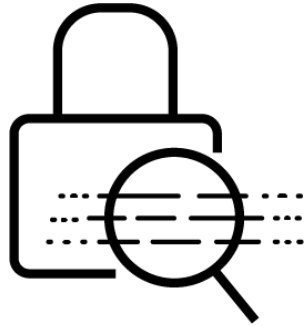
Instructor: Brian Krause

Announcements

Using Decryption to Block Threats in Encrypted Traffic

Some slides were adapted from Palo-Alto for use in this class. Content is a mesh between vendor provided information and instructor comments.

USING DECRYPTION TO BLOCK THREATS IN ENCRYPTED TRAFFIC



YOU CAN'T ANALYZE WHAT YOU CAN'T SEE

- SSL/TLS review
- Certificate management
- SSL/TLS decryption
- SSH decryption
- Other decryption methods and features

Learning Objectives

After you complete this module, you should be able to:

- Review fundamental SSL concepts and operation
- Create and manage certificates using the web interface
- Configure SSL/TLS forward proxy decryption
- Configure SSL/TLS inbound inspection decryption
- Prevent decryption for specific traffic
- View information and troubleshoot SSL/TLS issues using the CLI and logs
- Identify decryption configuration considerations
- Configure SSH decryption
- List other available decryption methods





SSL/TLS review

Certificate management

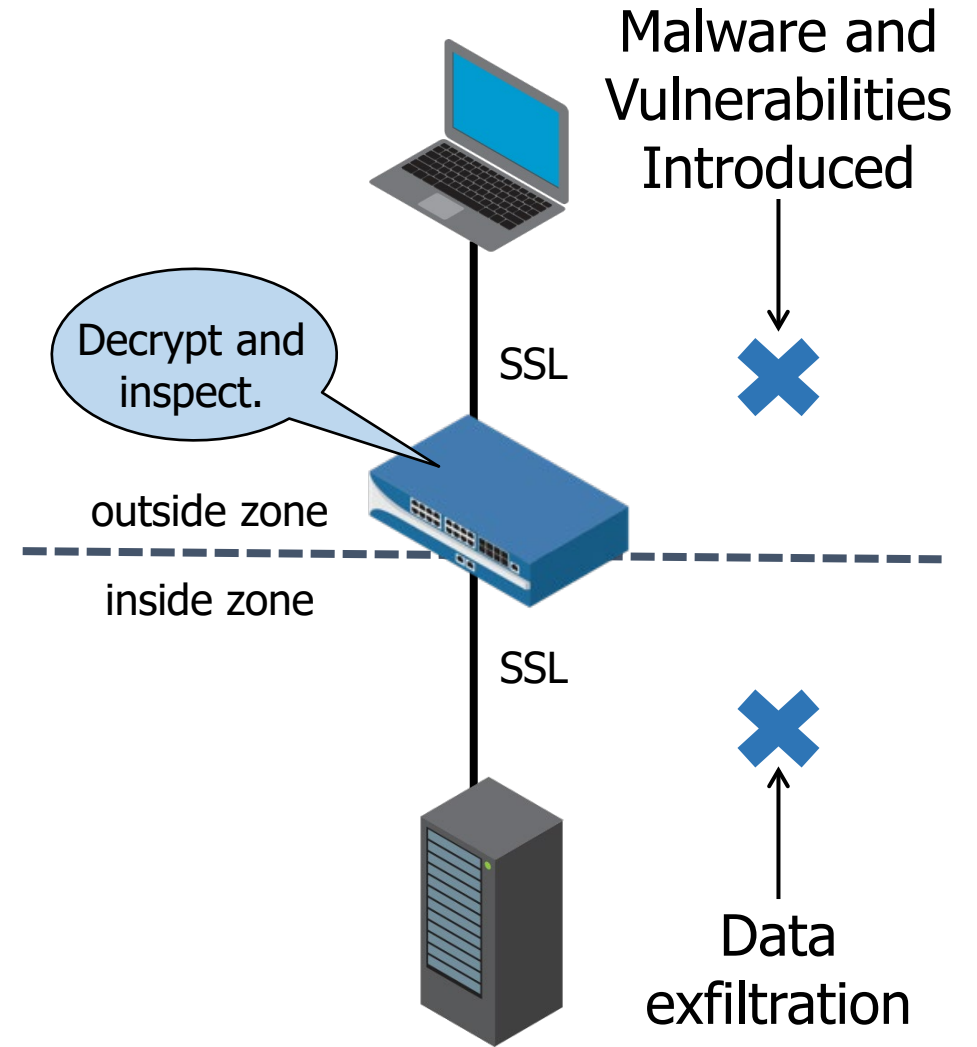
SSL/TLS decryption

SSH decryption

Other decryption methods and features

Importance of SSL/TLS

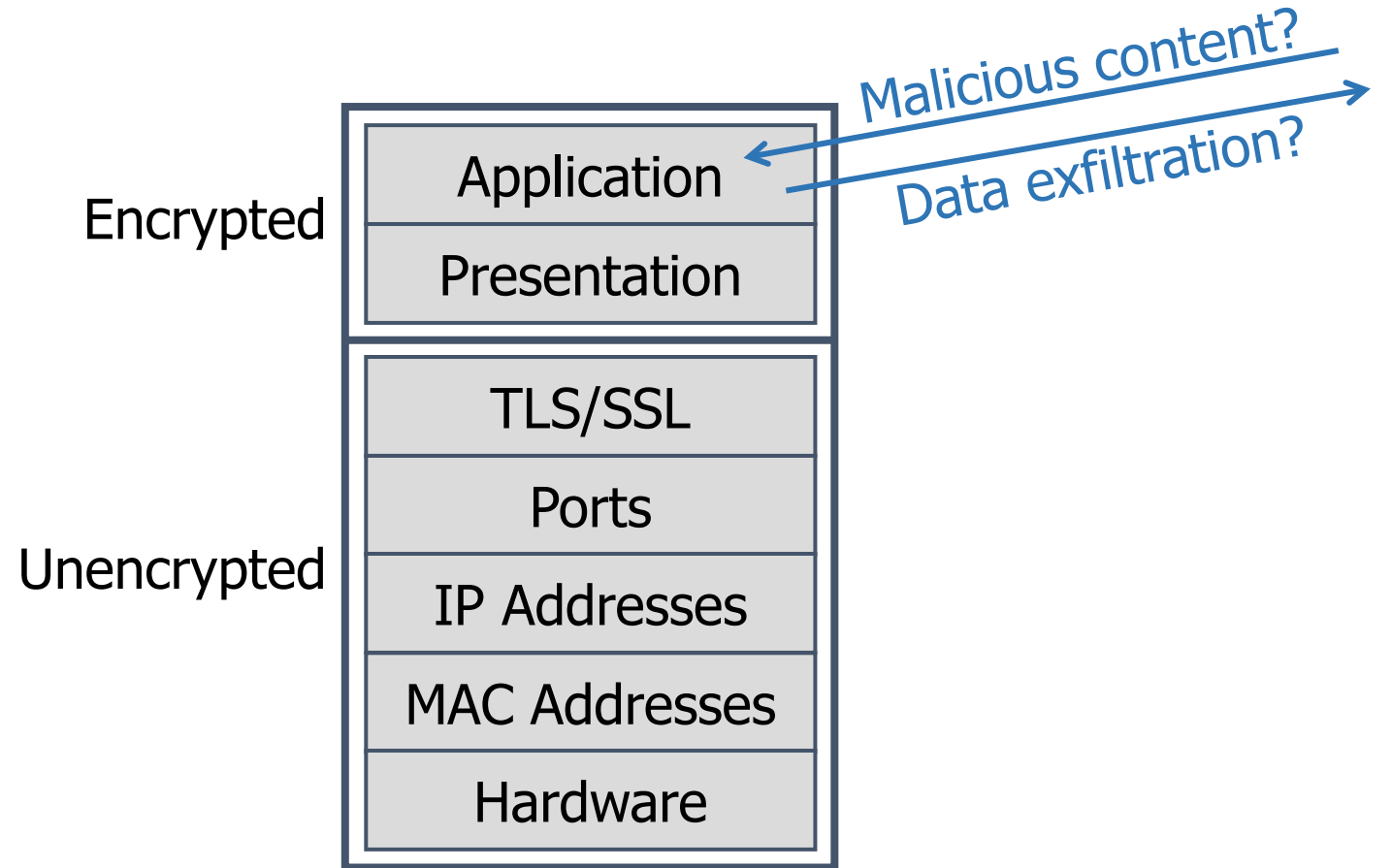
- SSL/TLS secures network communication across a shared network:
 - Encrypts for data privacy
 - Uses hashes for data integrity
 - Uses certificates for authentication
- SSL/TLS decryption helps to prevent:
 - Malware introduction
 - Data exfiltration



For a Tech Doc about this topic, log into Live and search for "Decryption Overview"

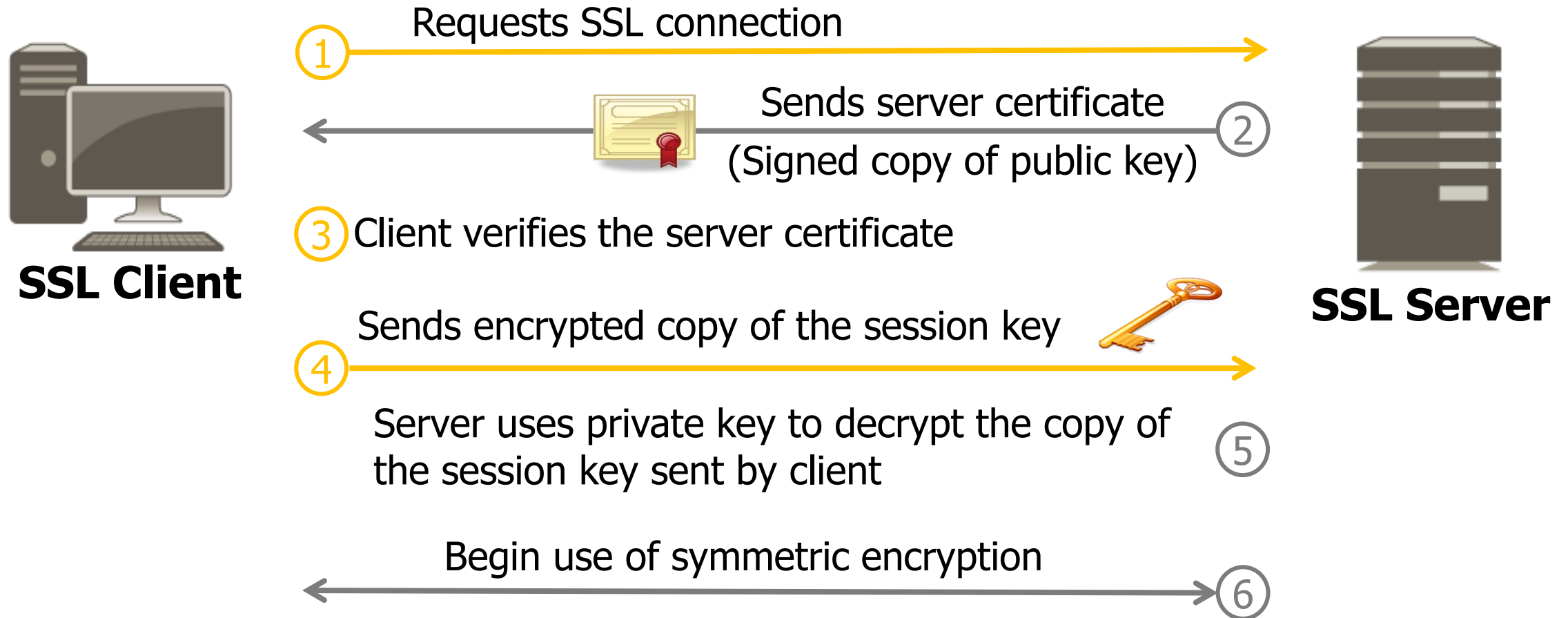
Why Decrypt Network Traffic?

- Most web traffic is encrypted.
- Palo Alto Networks firewalls can decrypt:
 - SSL/TLS inbound and outbound traffic
 - SSHv2



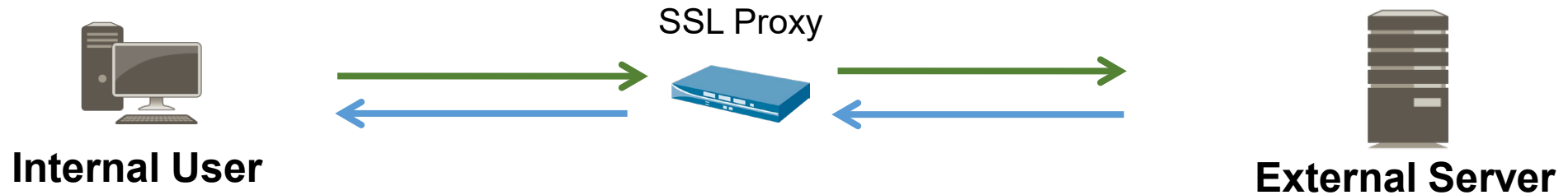
SSL/TLS Operation Review

SSL/TLS uses digital certificates to validate identity.

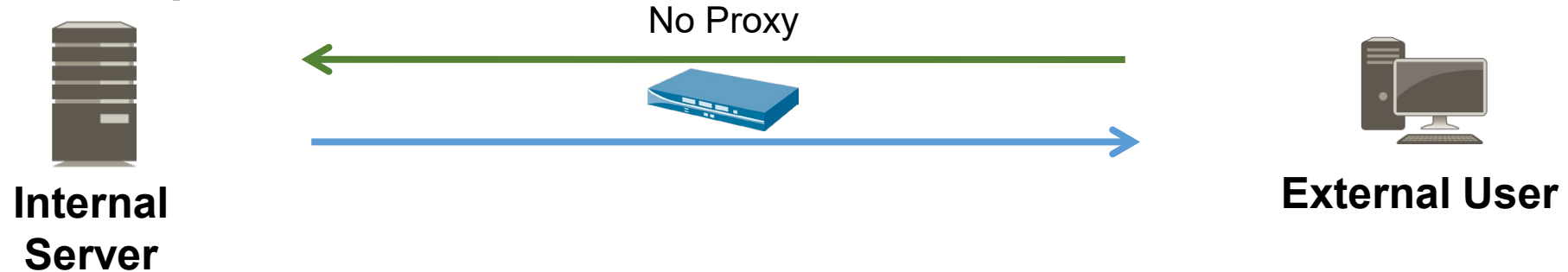


Firewall Decryption Types

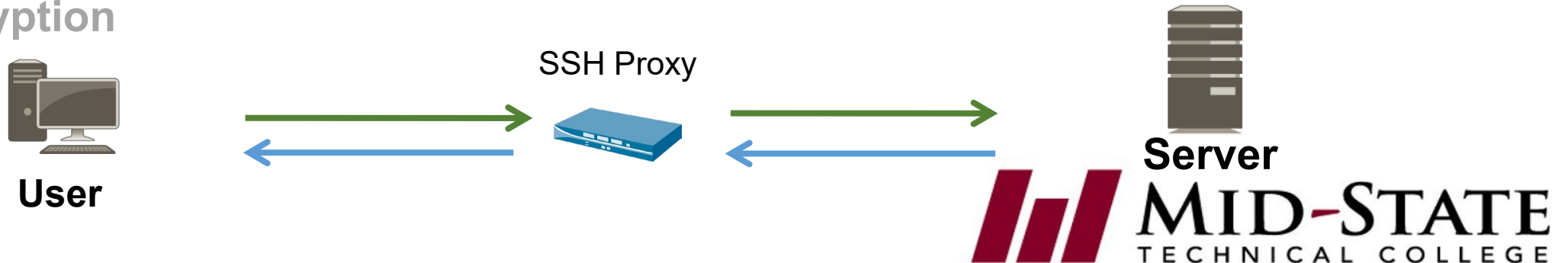
SSL Forward Proxy (Outbound)



SSL Inbound Inspection



SSH Decryption



SSL/TLS review



Certificate management

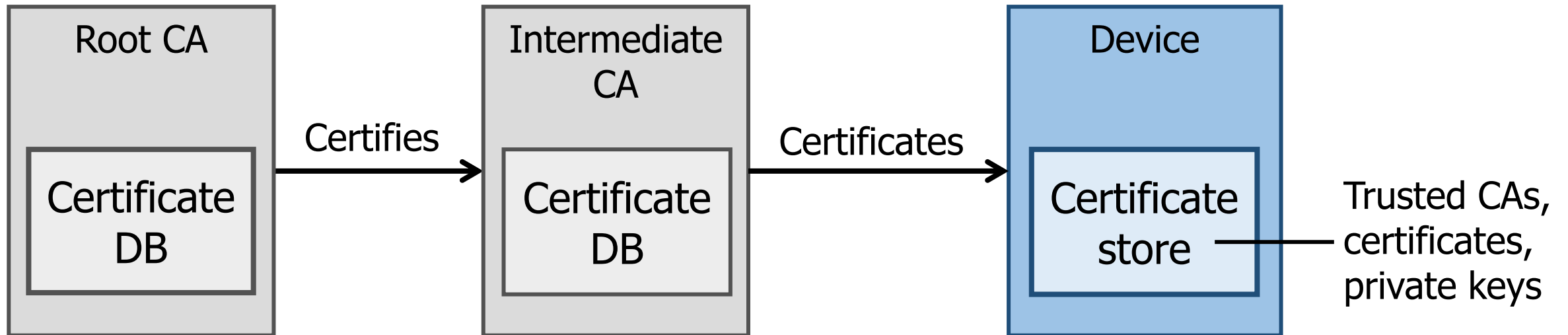
SSL/TLS decryption

SSH decryption

Other decryption methods and features

Public Key Infrastructure (PKI)

- Solves the problem of secure identification of public keys
- Uses digital certificates to verify public key owners
- Typical PKI components:



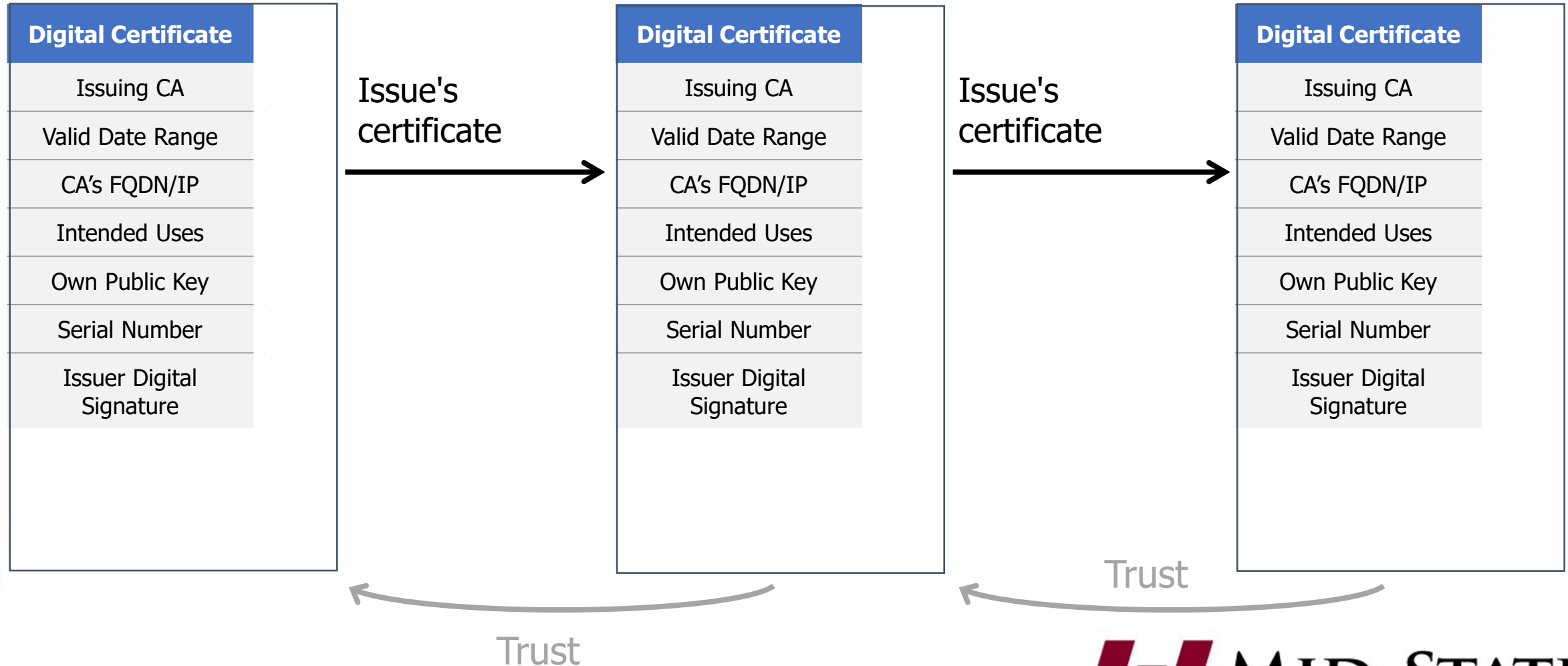
For a Tech Doc about this topic, log into Live and search for "Develop a PKI Rollout Plan"

Certificate Chain of Trust

Root CA

Intermediate CA

Device



Certificate Management in the Web Interface

Device > Certificate Management > Certificates

Device Certificates Default Trusted Certificate Authorities									
2 items → ×									
☐	NAME	SUBJECT	ISSUER	CA	KEY	EXPIRES	STATUS	ALGORITHM	USAGE
☐	FW-CA-Cert	C = US, O = Palo Alto N...	C = US, O = Palo Alto ...	☒	☒	Jul 24 00:36:21 202...	valid	RSA	Trusted Root CA Certificate
☐	Forward-Trust-Cert	C = US, O = Palo Alto N...	C = US, O = Palo Alto ...	☒	☒	Jul 24 00:37:23 202...	valid	RSA	Forward Trust Certificate
☐	Forward-Untrust-Cert	C = US, O = Palo Alto N...	C = US, O = Palo Alto ...	☒	☒	Jul 24 00:38:25 202...	valid	RSA	Forward Untrust Certificate
Delete Revoke Renew Import Generate Export Certificate Import HA Key Export HA Key PDF/CSV									

Types of operations:

- Generate certificates
- View certificates
- Modify certificate use
- Import and export certificates
- Delete certificates
- Renew and revoke certificates

Certificate Hierarchy

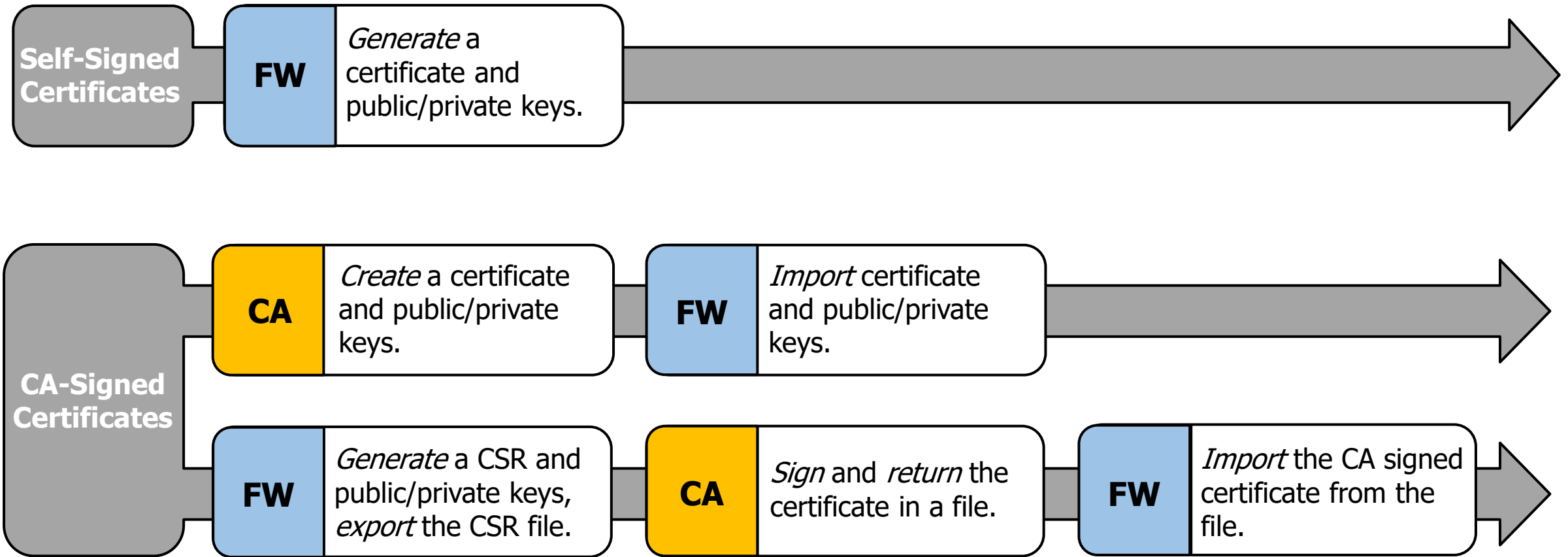
Device > Certificate Management > Certificates

Device Certificates Default Trusted Certificate Authorities									
Q 5 items → X									
☐	NAME	SUBJECT	ISSUER	CA	KEY	EXPIRES	STATUS	ALGORITHM	USAGE
☐	▼ FW-CA-Cert	C = US, O = Palo Alto N...	C = US, O = Palo Alto ...	☒	☒	Jul 24 00:36:21 202...	valid	RSA	Trusted Root CA Certificate
☐	Forward-Trust-Cert	C = US, O = Palo Alto N...	C = US, O = Palo Alto ...	☒	☒	Jul 24 00:37:23 202...	valid	RSA	Forward Trust Certificate
☐	GP-Portal	C = US, O = Palo Alto N...	C = US, O = Palo Alto ...	☐	☒	Jul 24 00:45:35 202...	valid	RSA	
☐	GP-Gateway	C = US, O = Palo Alto N...	C = US, O = Palo Alto ...	☐	☒	Jul 24 00:46:26 202...	valid	RSA	
☐	WebUI-Cert	C = US, O = Palo Alto N...	C = US, O = Palo Alto ...	☐	☒	Jul 24 00:47:33 202...	valid	RSA	
☐	Forward-Untrust-Cert	C = US, O = Palo Alto N...	C = US, O = Palo Alto ...	☒	☒	Jul 24 00:38:25 202...	valid	RSA	Forward Untrust Certificate
⊖ Delete ↺ Revoke ↻ Renew ⬇ Import 🔧 Generate 📄 Export Certificate ⬇ Import HA Key 📄 Export HA Key 📄 PDF/CSV									

Nested hierarchy visually illustrates the certificate chain of trust.

Certificate Creation Overview

Methods to obtain required certificates and public/private keys:



For a Tech Doc about this topic, log into Live and search for "Certificate Deployment"

Generate a Self-Signed Certificate

Device > Certificate Management > Certificates > Add

Generate Certificate

Certificate Type ☒ Local ☐ SCEP

Certificate Name Forward-Untrusted-Cert

Common Name Not Trusted
IP or FQDN to appear on the certificate

Signed By

☒ Certificate Authority
☐ Block Private Key Export

OCSP Responder

Cryptographic Settings

Algorithm RSA

Number of Bits 2048

Digest sha256

Expiration (days) 365

Certificate Attributes

☐	TYPE	VALUE
☐	Country = "C" from "Subject" field	US
☐	Organization = "O" from "Subject" field	Palo Alto Networks

+ Add - Delete

Leave blank to create a self-signed certificate.

Select to create a CA certificate.

- Configure a self-signed certificate:
 - The **Signed By** field must be blank.
- **Generate** creates a certificate and public/private key pair.

For a Tech Doc about this topic, log into Live and search for "Create a Self-Signed Root CA Certificate."

Import a CA Certificate

- Use an internal CA to create:
 - Firewall CA certificate
 - Public/private key pair
- Use **Device > Certificate Management > Certificates > Import**.
- Complete the form and click **OK**.
- Imports certificate and public/private keys into the firewall.

Delete Revoke Renew **Import** Generate

Import Certificate ⓘ

Certificate Type ☒ Local ☐ SCEP ☐ Cloud

Certificate Name

Certificate File [Browse...](#)

File Format ▼

☐ Private key resides on Hardware Security Module

☒ Import Private Key

☒ Block Private Key Export

This option will permanently block export of private key for this certificate

Key File [Browse...](#)

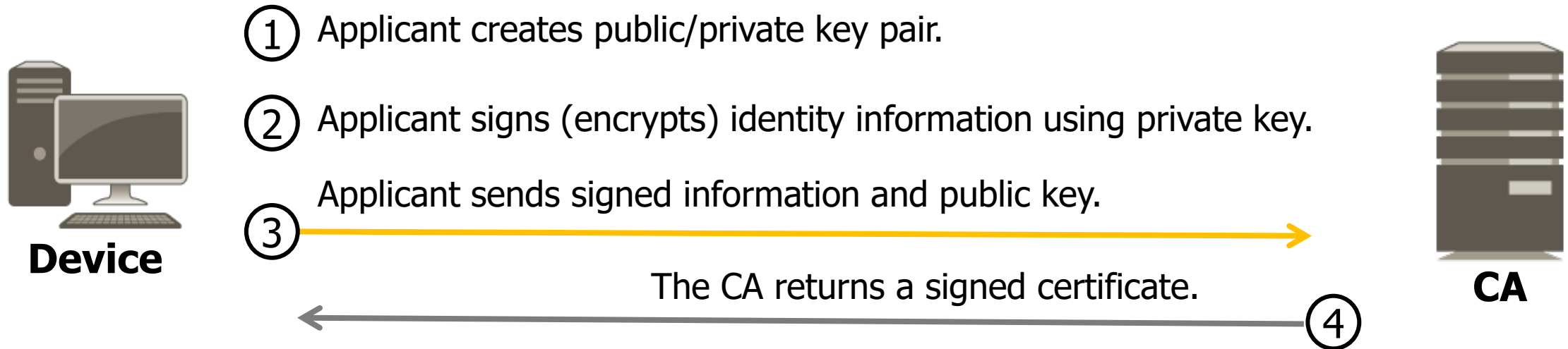
Passphrase

Confirm Passphrase

For a Tech Doc about this topic, log into Live and search for "Import a Certificate and Private Key"

Certificate Signing Request (CSR)

Message sent to CA to acquire a certificate



Advantages:

- Device is part of PKI and benefactor of chain of trust.
- Private key never leaves device.

For a Tech Doc about this topic, log into Live and search for "Obtain a Certificate from an External CA"

Generate a CSR for the CA-Signed Certificate

Device > Certificate Management > Certificates > Add

The screenshot shows the 'Generate Certificate' form. Two callout boxes provide additional information:

- CA will make this a certificate authority.** This points to the 'Signed By' dropdown menu, which is set to 'External Authority (CSR)'. Below this dropdown are two checkboxes: 'Certificate Authority' (unchecked) and 'Block Private Key Export' (unchecked).
- Blocks the export of private keys that are generated or imported** This points to the 'Block Private Key Export' checkbox.

Cryptographic Settings

- Certificate Type: ☒ Local
- Certificate Name: Forward-Trusted-Cert
- Common Name: Firewall-A
- IP or FQDN to appear on the cert: (empty)
- Signed By: External Authority (CSR)
- ☐ Certificate Authority
- ☐ Block Private Key Export
- OCSP Responder: (empty)
- Algorithm: RSA
- Number of Bits: 2048
- Digest: sha256
- Expiration (days): 365

Certificate Attributes

NAME	VALUE
Country = "C" from "Subject" field	US
Organization = "O" from "Subject" field	Palo Alto Networks

+ Add - Delete

- Generate a CSR.
- Export the CSR file to the signing CA.
- CA returns the signed intermediate CA certificate.
- Import the signed intermediate CA certificate to the firewall.

The screenshot shows the 'Device Certificates' table. Two callout boxes highlight specific actions:

- Import** (with a download icon): This button is used to import a certificate file into the device.
- Export Certificate** (with an upload icon): This button is used to export a certificate file from the device.

NAME	SUBJECT	ISSUER	CA	KEY	EXPIRES	STATUS
☐ FW-CA-Cert	C = US, O = Palo Alto...	C = US, O = Palo Alto...	☒	☒	Jul 23 20:22:52 202...	valid
☒ Forward-Trusted-Cert	Firewall-A		☐	☒		pending
☐ Forward-Untrusted-Cert	C = US, O = Palo Alto...	C = US, O = Palo Alto...	☒	☒	Jul 23 20:26:09 202...	valid

Buttons: Delete, Revoke, Renew, Import, Generate, Export Certificate, Import HA Key, Export HA Key, PDF/CSV

.pem or .cer
Base64 encoding required

Certificate Checking and Revocation

Determine certificate chain of trust.



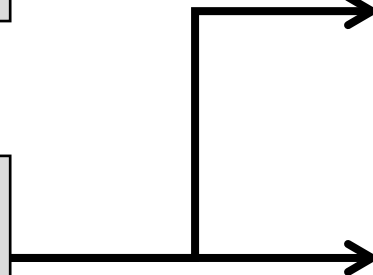
Validate each certificate in the chain:

- Signature valid?
- Date range valid?
- Malformed or corrupt?



Check the revocation status of each certificate.

- Reasons to revoke certificates:
 - Private key compromised
 - Hostname or username of owner changed
 - Host decommissioned; user left company
 - Counterfeit key found
- Possible status returned:
 - Current, expired, unknown, unavailable



Online Certificate Status Protocol (OCSP)

Certificate revocation list (CRL)

Processing order



Configuring SSL Decryption Certificate Revocation Checking

Device > Setup > Session > Certificate Revocation Checking

Certificate Revocation Checking

CRL

☒ Enable
Use CRL to check certificate status

Receive Timeout (sec) 5

OCSP

☒ Enable
Use OCSP to check certificate status

Receive Timeout (sec) 5

Certificate Status Timeout (sec) 5
Certificate CRL status query timeout value

You can select either or both to enable certificate revocation checking.

SSL/TLS review

Certificate management



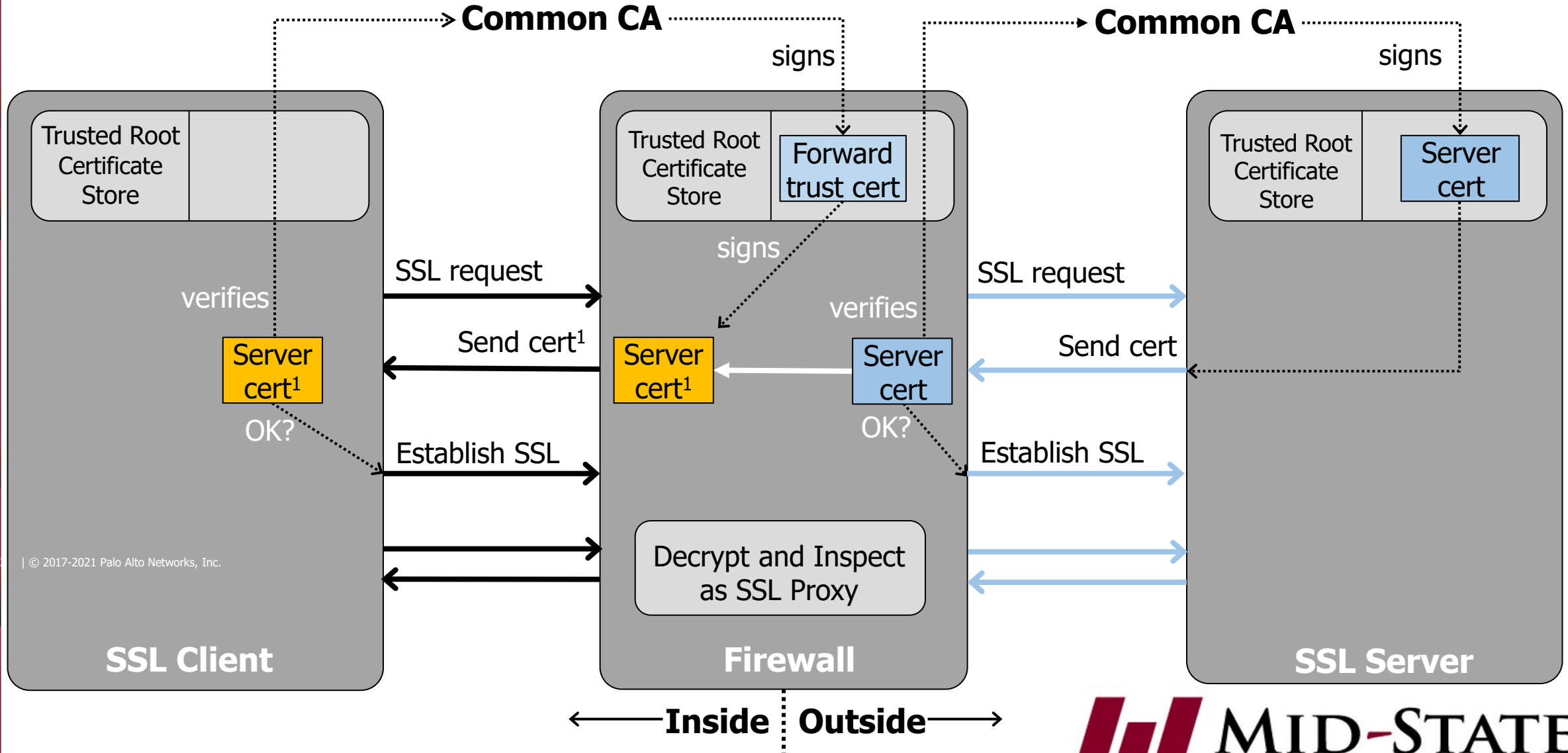
SSL/TLS decryption

SSH decryption

Other decryption methods and features

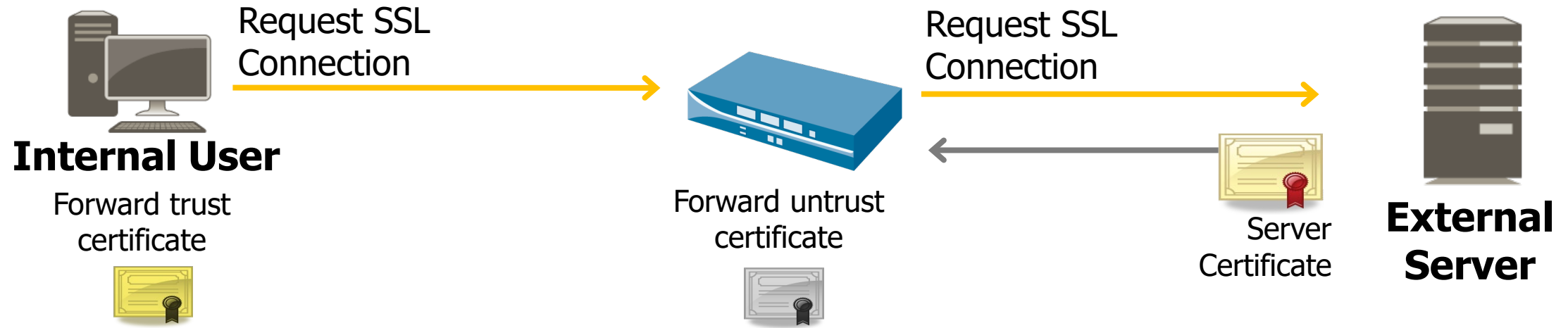
SSL Forward Proxy Review

For a Tech Doc about this topic, log into Live and search for "SSL Forward Proxy"



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Forward Trust and Forward Untrust Certificates



← Firewall signs a copy of a server certificate with a forward trust certificate.



If server certificate is **trusted**



← Firewall signs a copy of a server certificate with a forward untrust certificate.



If server certificate is **untrusted**

Configure a Forward Trust Certificate

Device > Certificate Management > Certificates

Device Certificates | Default Trusted Certificate Authorities

Search

☐	NAME	SUBJECT
☐	Forward-Untrusted-Cert	C = US, O
☐	Forwrd-Trust-Cert	C = US, O

Must be a valid CA certificate and trusted by the client

Certificate information

Name: Forwrd-Trust-Cert

Subject: /C=US/O=Palo Alto Networks/CN=Firewall-A

Issuer: /C=US/O=Palo Alto Networks/CN=Firewall-A

Not Valid Before: Jul 23 21:24:33 2020 GMT

Not Valid After: Jul 23 21:24:33 2021 GMT

Algorithm: RSA

☒ Certificate Authority

☒ Forward Trust Certificate

☐ Forward Untrust Certificate

☐ Trusted Root CA

Select.

Configure a Forward Untrust Certificate

Device > Certificate Management > Certificates

Device Certificates | Default Trusted Certificate Authorities

Search

☐	NAME	SUBJECT
☐	Forward-Untrusted-Cert	C = US, O
☐	Forward-Trust-Cert	C = US, O

Must be a CA certificate and *not* trusted by the client

Certificate information

Name: Forward-Untrusted-Cert

Subject: /C=US/O=Palo Alto Networks/CN=Not Trusted

Issuer: /C=US/O=Palo Alto Networks/CN=Not Trusted

Not Valid Before: Jul 23 20:26:09 2020 GMT

Not Valid After: Jul 23 20:26:09 2021 GMT

Algorithm: RSA

☒ Certificate Authority

☐ Forward Trust Certificate

☒ Forward Untrust Certificate

☐ Trusted Root CA

Select.

Renew an SSL Forward Untrust Certificate

Renews SSL forward untrust certificate issued by the firewall

Device > Certificate Management > Certificates

Device Certificates | Default Trusted Certificate Authorities

☐	NAME	SUBJECT	ISSUER	CA	KEY	EXPIRES	STATUS	ALGORITHM
☐	Forward-Untrusted-Cert	C = US, O = Pal...	C = US, O = Pal...	☒	☒	Jul 25 21:36:5...	valid	RSA
☐	Forwrd-Trust-Cert	C = US, O = Pal...	C = US, O = Pal...	☒	☒	Jul 25 21:37:1...	valid	RSA
☐	Web-Server1-Cert	C = US, O = Pal...	C = US, O = Pal...	☒	☒	Jul 23 21:35:2...	expired	RSA

Delete Revoke **Renew** Import Generate Export Cert

Renew Certificate - Web-Server1-Cert

New Expiration Interval (days)

OK Cancel

Configure SSL Forward Proxy Decryption Policy

Policies > Decryption

Decryption Policy Rule

General | Source | Destination | **Service/URL Category** | Options

Match conditions

select

☐ SERVICE ^

☐ service-https

☐ Service-62443

Consider SSL traffic on non-default ports.

☐ URL CATEGORY ^

+ Add - Delete

- Use rule fields to control what is decrypted.
- Phase in decryption to minimize user issues.
- Decryption subject to legal and privacy concerns (health, HR, finance, etc.)
- Create a Security policy rule to allow the traffic.

Decryption Policy Rule

Source | Destination | Service/URL Category | **Options**

Action ☐ No Decrypt ☒ Decrypt

Type SSL Forward Proxy

Decryption Profile Outbound-Traffic

Configures whether matched traffic is decrypted

SSL Forward Proxy

SSH Proxy

SSL Inbound Inspection

Forward Proxy Decryption Profile

Objects > Decryption > Decryption Profile

The screenshot shows the 'Decryption Profile' configuration for 'Outbound-Traffic'. The 'SSL Decryption' tab is active, with sub-tabs for 'SSL Forward Proxy', 'SSL Inbound Inspection', and 'SSL Protocol Settings'. Under 'Server Certificate Verification', several options are checked: 'Block sessions with expired certificates', 'Block sessions with untrusted issuers', 'Block sessions with unknown certificate status', and 'Block sessions on certificate status check timeout'. Under 'Unsupported Mode Checks', 'Block sessions with unsupported versions' and 'Block sessions with unsupported cipher suites' are checked. Under 'Failure Checks', 'Block sessions if resources not available', 'Block sessions if HSM not available', and 'Block downgrade on no resource' are unchecked. Under 'Client Extension', 'Strip ALPN' is unchecked.

- An SSL Forward Proxy policy rule specifies what to decrypt.
- An attached Decryption Profile specifies additional certificate and protocol checks.

Policies > Decryption

The screenshot shows the 'Decryption Policy Rule' configuration for 'Outbound-Traffic'. The 'Options' tab is active. The 'Action' is set to 'Decrypt' (selected with a radio button). The 'Type' is 'SSL Forward Proxy'. The 'Decryption Profile' is 'Outbound-Traffic'. A callout box points to the 'Decryption Profile' field with the text 'Apply profile to policy.'

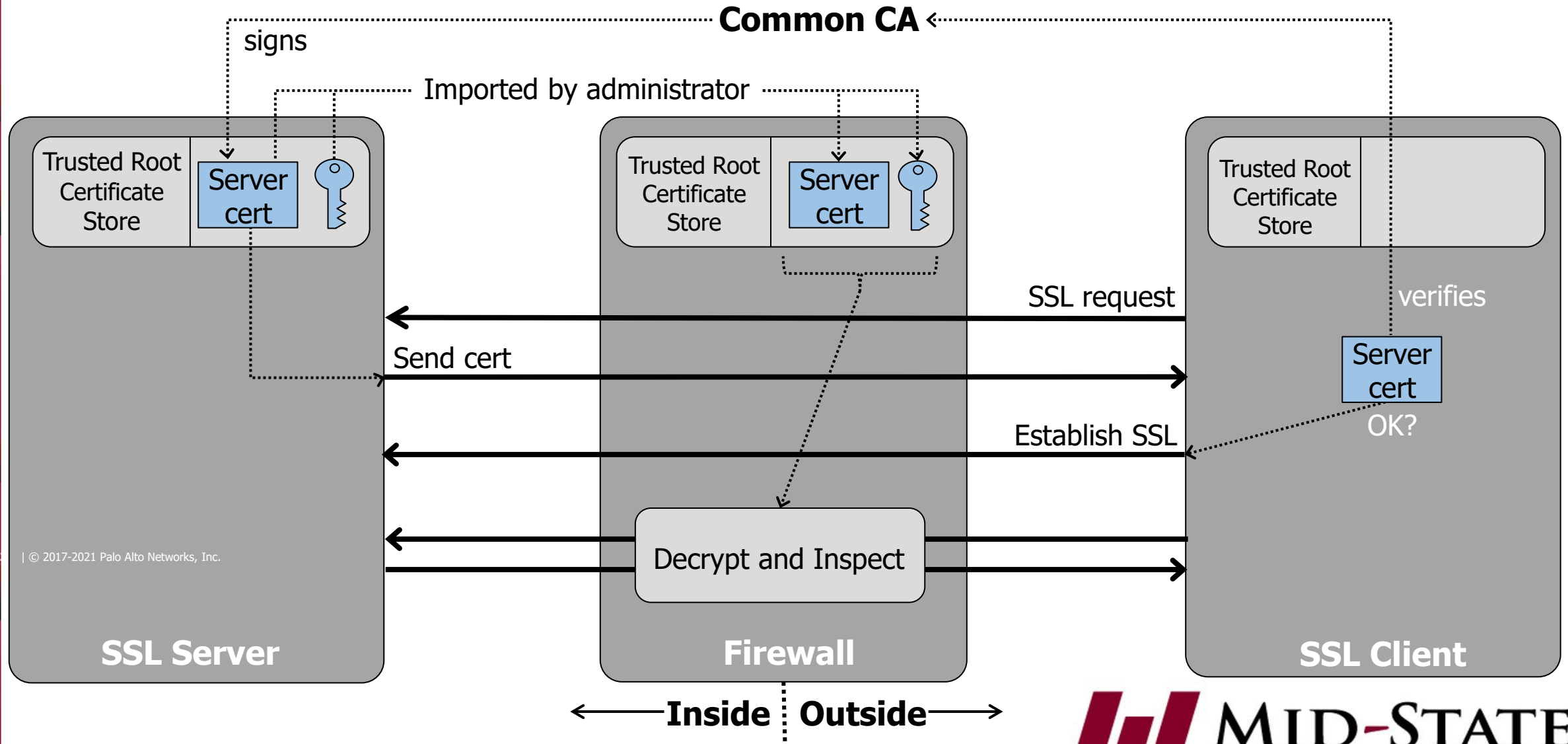
The screenshot shows the 'Decryption Profile' configuration for 'Outbound-Traffic' with the 'SSL Protocol Settings' sub-tab active. The 'Min Version' is 'TLSv1.0' and the 'Max Version' is 'Max'. A dropdown menu is open for 'Min Version', showing options: 'SSLv3.0', 'TLSv1.0', 'TLSv1.1', 'TLSv1.2', and 'TLSv1.3'. A callout box points to 'TLSv1.3' with the text 'Added support for TLSv1.3'. Other settings include 'Key Exchange Algorithms' (RSA, DHE, ECDSA), 'Encryption Algorithms' (3DES, RC4, AES128-CBC, AES256-CBC, AES256-GCM, POLY1305), and 'Authentication Algorithms' (MD5, SHA1, SHA256, SHA384).

Some older algorithms no longer are considered fully secure.

Added support for TLSv1.3

SSL Inbound Inspection Review

For a Tech Doc about this topic, log into Live and search for "Inbound Inspection"



Import Server Certificate and Private Key

Import the internal server certificate and private key into the firewall.

Import Certificate ?

Certificate Type

☒ Local ☐ SCEP ☐ Cloud

Certificate Name

Server-Cert

Certificate File

C:\fakepath\Server-Cert.p12 [Browse...](#)

File Format

Encrypted Private Key and Certificate (PKCS12) ▼

☐ Block Private Key Export

This option will permanently block export of private key

Passphrase

●●●●●●●●

Confirm Passphrase

●●●●●●●●

Base64 Encoded Certificate (PEM)

Encrypted Private Key and Certificate (PKCS12)

For a Tech Doc about this topic, log into Live and search for "Configure SSL Inbound Inspection"

Configure an SSL Inbound Inspection Policy

- An SSL Inbound Inspection policy rule specifies what to inspect.
- An attached profile specifies additional protocol and firewall resource checks.
- Create a Security policy rule that allows traffic.

Policies > Decryption > Add

Decryption Policy Rule

General | Source | Destination | Service/URL Category | **Options**

Action ☐ No Decrypt ☒ Decrypt

Type SSL Inbound Inspection

Certificates

	CERTIFICATES ^
☐	Server-Cert
☐	trusted-cert

+ Add - Delete

Decryption Profile Inbound-Traffic

Imported certificate(s)

Configure an Inbound Inspection Decryption Profile

Objects > Decryption > Decryption Profile > Add

Decryption Profile

Name

Inbound-Traffic

SSL Decryption

No Decryption

SSH Proxy

SSL Forward Proxy

SSL Inbound Inspection

SSL Protocol Settings

Unsupported Mode Checks

☒ Block sessions with unsupported versions

☒ Block sessions with unsupported cipher suites

Failure Checks

☐ Block sessions if resources not available

☐ Block sessions if HSM not available

☐ Block downgrade on no resource

Note: For unsupported modes and failures, the session information is cached for 12 hours, so future sessions between the same host and server pair are not decrypted. Check boxes to block those sessions instead.

Default setting allows encrypted traffic if firewall is too busy.

For a Tech Doc about this topic, log into Live and search for "Create a Decryption Profile"

For a Tech Doc about this topic, log into Live and search for "Decryption Exclusions"

SSL Decryption Exclusion ?

Hostname

Description

☒ Exclude
Note: check to exclude entry from decryption

Globally disables decryption regardless of Decryption policy

+ Add - Delete Clone Enable Disable Show obsoletes Excluded Common Names and SNIs PDF/CSV Show Local Exclusion Cache

- Websites with known decryption problems are pre-populated on the list:
 - Exclusion list updated via content updates
- You can add websites to the exclusion list.

No Decryption

Even if the Decryption policy rule action is “no-decrypt,” the Decryption Profile can be configured to block sessions with expired or untrusted certificates.

Policies > Decryption

ACTION	TYPE	DECRYPTION PROFILE
no-decrypt	ssl-forward-proxy	No-Decryption
no-decrypt	ssl-forward-proxy	No-Decryption

Objects > Decryption Profile > Add

Decryption Profile

Name

No-Decryption

SSL Decryption

No Decryption

SSH Proxy

Server Certificate Verification

☒

Block sessions with expired certificates

☒

Block sessions with untrusted issuers

Note: For unsupported modes and failures, the session information is cached for 12 hours, boxes to block those sessions instead.

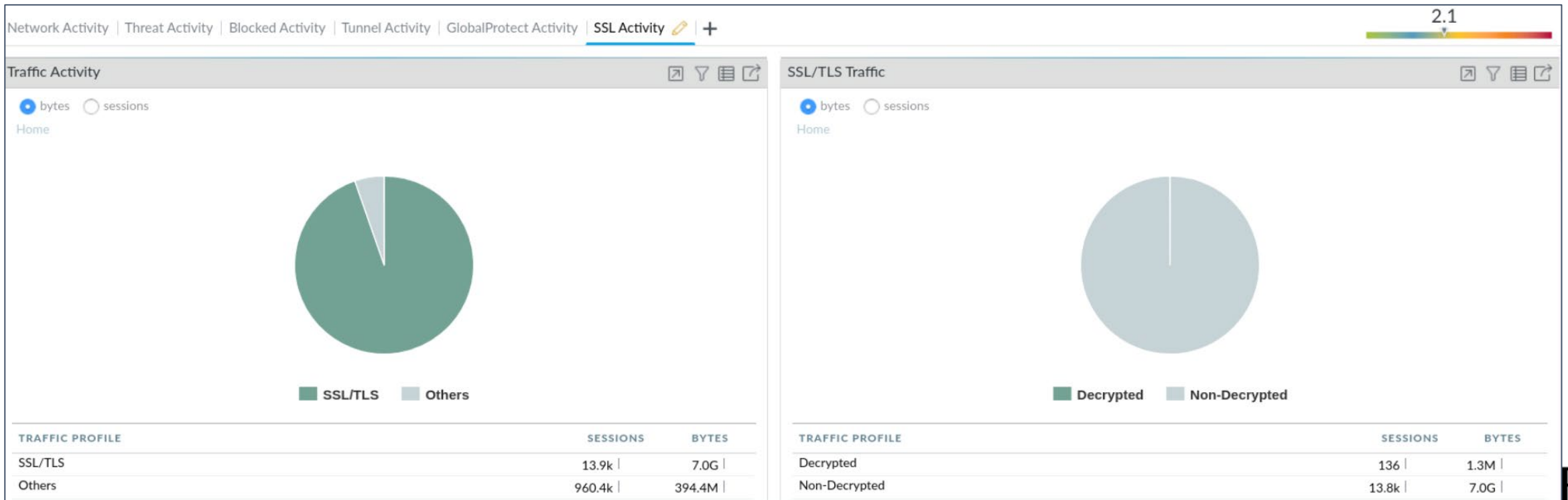
For a Tech Doc about this topic, log into Live and search for “Create a Policy-Based Decryption Exception”

SSL Decryption Troubleshooting

Monitor > Logs > Decryption

	RECEIVE TIME	APPLICATION	POLICY NAME	SOURCE ZONE	DESTINA... ZONE	PROXY TYPE	SOURCE ADDRESS	DESTINATION ADDRESS	ROOT STATUS	TLS VERSION	KEY EXCHANGE	ENCRYPTION ALGORITHM
	07/23 18:05:00	web-browsing	Decrypts_User_Traffic	Users_Net	Internet	Forward	192.168.1.20	172.217.12.42	trusted	TLS1.2	ECDHE	AES_128_GCM
	07/23 17:20:02	web-browsing	Decrypts_User_Traffic	Users_Net	Internet	Forward	192.168.1.20	172.217.1.237	trusted	TLS1.2	ECDHE	AES_128_GCM
	07/23 17:14:55	web-browsing	Decrypts_User_Traffic	Users_Net	Internet	Forward	192.168.1.20	216.58.193.131	trusted	TLS1.2	ECDHE	AES_128_GCM

ACC > SSL Activity

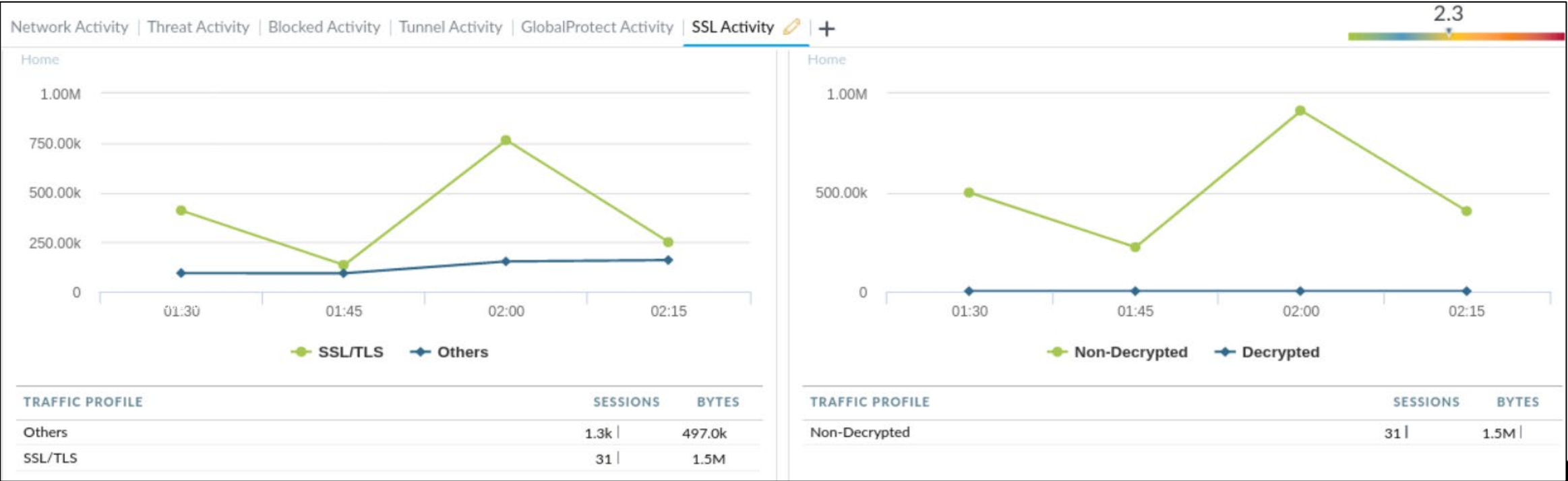


SSL Decryption Troubleshooting

Monitor > Logs > Decryption

	RECEIVE TIME	APPLICATION	POLICY NAME	SOURCE ZONE	DESTINA... ZONE	PROXY TYPE	SOURCE ADDRESS	DESTINATION ADDRESS	ROOT STATUS	TLS VERSION	KEY EXCHANGE	ENCRYPTION ALGORITHM
	07/23 18:05:00	web-browsing	Decrypts_User_Traffic	Users_Net	Internet	Forward	192.168.1.20	172.217.12.42	trusted	TLS1.2	ECDHE	AES_128_GCM
	07/23 17:20:02	web-browsing	Decrypts_User_Traffic	Users_Net	Internet	Forward	192.168.1.20	172.217.1.237	trusted	TLS1.2	ECDHE	AES_128_GCM
	07/23 17:14:55	web-browsing	Decrypts_User_Traffic	Users_Net	Internet	Forward	192.168.1.20	216.58.193.131	trusted	TLS1.2	ECDHE	AES_128_GCM

ACC > SSL Activity



Troubleshoot SSL Session Terminations

Monitor > Logs > Traffic

Session end log entries

	CRYPTED	TYPE	FROM ZONE	TO ZONE	SOURCE	DESTINATION	TO PORT	APPLICATION	ACTION	RULE
		end	Users_Net	Internet	192.168.1.252	34.96.84.34	443	web-browsing	allow	Users_to_Internet
		end	Users_Net	Internet	192.168.1.254	34.96.84.34	443	web-browsing	allow	Users_to_Internet
07/23 18:00:21	yes									Users_to_Internet
07/23 18:00:21										Users_to_Internet
07/23 18:00:21										Users_to_Internet
07/23 18:00:21										Users_to_Internet
07/23 18:00:21										Users_to_Internet
07/23 17:59:59										Users_to_Internet
07/23 17:59:59										Users_to_Internet

Add Log Filter

(session_end_reason eq decrypt-error)

Connector	Attribute	Operator	Value
and	SDWAN Cluster Type	equal	unknown
or	SDWAN Policy Name	not equal	decrypt-cert-validation
	SDWAN Site Name		decrypt-unsupported-param
	SDWAN Site Type		decrypt-error
	Session End Reason		split-tunnel
	Session ID		N/A

☐ Negate

Filter log for SSL-related errors.

Decryption in the Traffic Log

For a Tech Doc about this topic, log into Live and search for "Configure Decryption Logging"

Monitor > Logs > Traffic

The screenshot displays the 'Monitor > Logs > Traffic' interface. A table lists traffic logs with columns: RECEIVE TIME, DECRYPTED, TYPE, FROM ZONE, TO ZONE, SOURCE, DESTINATION, TO PORT, APPLICATION, ACTION, and RULE. The 'DECRYPTED' column for the second log entry (07/23 18:30:24) is highlighted with a black box. An arrow points from this box to the 'Detailed Log View' window.

Detailed Log View

Rule UUID	d6fede9b-9a48-419a-ac7f-fc0b83993fcb	Zone	Users_Net	Interface	ethernet1/2	Interface	ethernet1/1
Session End Reason	aged-out	NAT IP	203.0.113.20	NAT IP	34.96.84.34	NAT Port	443
Category	computer-and-internet-info	NAT Port	13290				
Device SN		X-Forwarded-For IP	0.0.0.0				
IP Protocol	tcp						
Log Action							
Generated Time	2020/07/23 18:30:24						
Start Time	2020/07/23 18:30:00						
Receive Time	2020/07/23 18:30:24						
Elapsed Time(sec)	10						
Tunnel Time	N/A						

Flags

Captive Portal	☐
Proxy Transaction	☐
Decrypted	☒
Packet Capture	☐
Client to Server	☐
Server to Client	☐

SSL/TLS review

Certificate management

SSL/TLS decryption

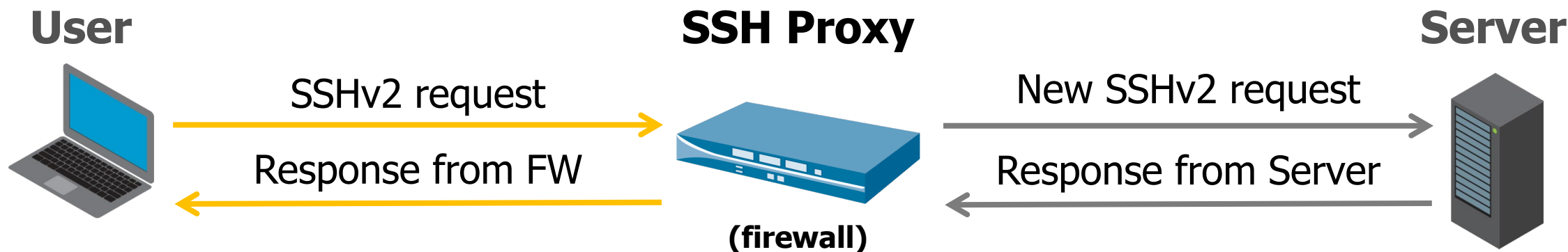
SSH decryption



Other decryption methods and features

SSH Decryption

For a Tech Doc about this topic, log into Live and search for "Configure SSH Proxy"

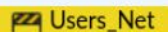


- Decrypts and inspects SSHv2 traffic (to detect SSH-tunneled applications).
- Unsupported with SSH key passwordless login.
- Uses an automatically generated key to decrypt or encrypt traffic.
- All traffic is identified as either *ssh* or *ssh-tunnel*:
 - Control traffic using Security policy rules.

SSH Traffic and the Security Policy

Add rules to control ssh and ssh-tunnel traffic.

Policies > Security > Add

	NAME	TAGS	TYPE	Source			Destination		APPLICATION	SERVICE	URL CATEGORY	ACTION
				ZONE	ADDRESS	USER	ZONE	ADDRESS				
1	Bad-SSH-Traffic	Users_Net	universal	 Users_Net	 165.35.13.6	any	 Extranet	any	 ssh	any	any	 Deny
2	Block-SSH-Tunnels	Users_Net	universal	 Users_Net	any	any	 Extranet	any	 ssh-tunnel	any	any	 Deny
3	Good-SSH-Traffic	Users_Net	universal	 Users_Net	any	any	 Extranet	any	 ssh	 application-default	any	 Allow

SSL/TLS review

Certificate management

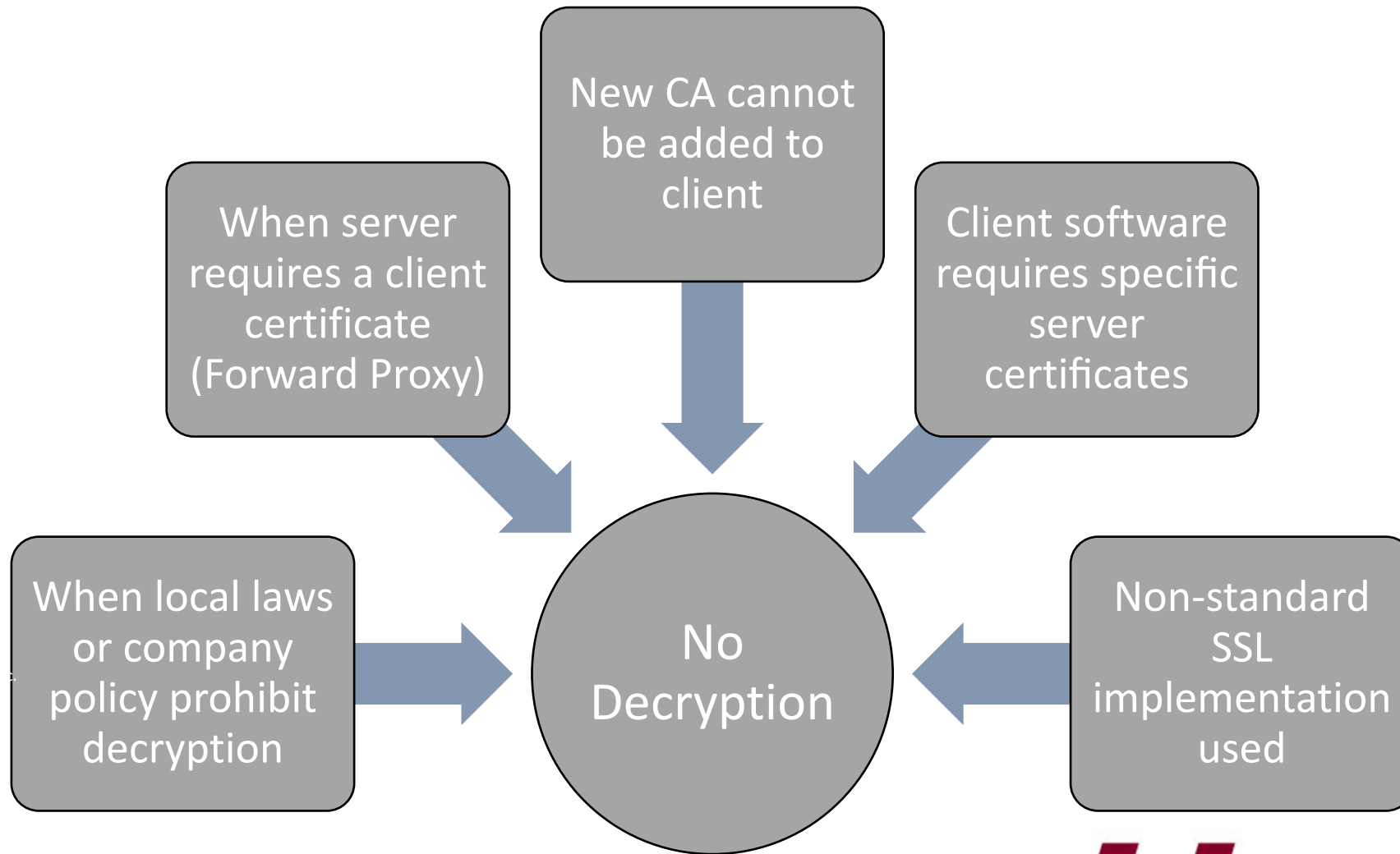
SSL/TLS decryption

SSH decryption

Other decryption methods and features

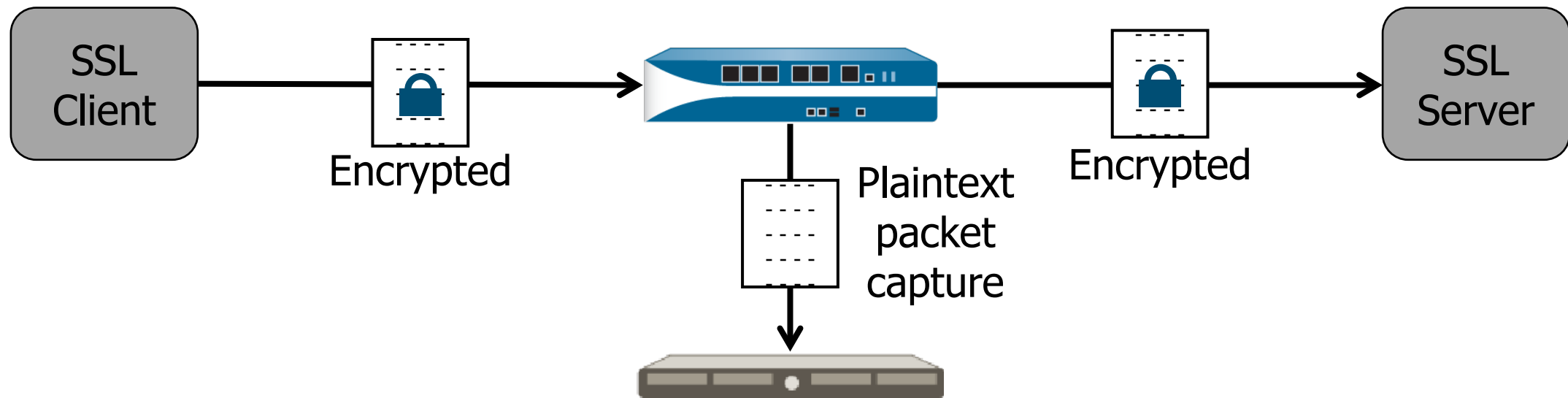


Reasons to Not Configure SSL Decryption



Decryption Port Mirroring

- Exports decrypted flows out of a dedicated interface on the firewall.
- Use cases include data loss prevention (DLP) and network forensics.
- Requires a free license.



For a Tech Doc about this topic, log into Live and search for "Decryption Mirroring"

Network Packet Broker

For a Tech Doc about this topic, log into Live and search for "Objects > Packet Broker Profile"

- Replaces Decryption Broker
- Filtered traffic is forwarded to security chain
- Identifies non-decrypted TLS, decrypted TLS, and non-TLS (TCP and UDP) traffic to forward

Objects > Packet Broker Profile

Packet Broker Profile ?

Name

Remote Users Security Chain

Description

Inspect traffic from remote users

General

Security Chains

Health Monitor

Security Chain Type

Routed (Layer 3)

Flow Direction

☐ Unidirectional

☒ Bidirectional

Client-to-Server flow via Interface #1
Server-to-Client flow via Interface #2

Interface #1

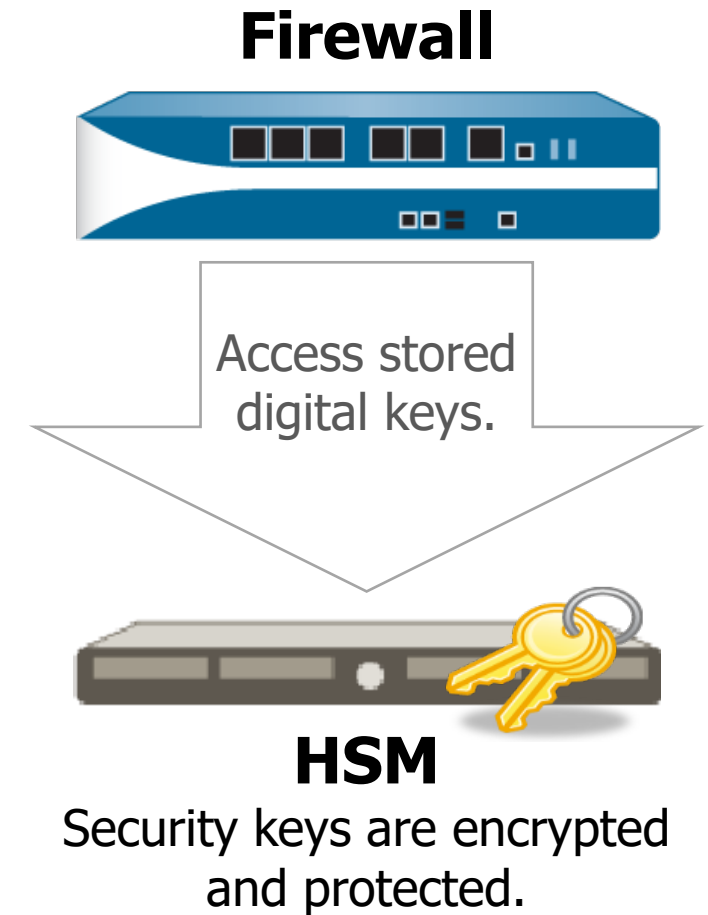
ethernet1/10

Interface #2

ethernet1/11

Hardware Security Modules (HSMs)

- Physical device designed to safeguard digital keys
- Generates, stores, and manages digital keys
- Used by firewall:
 - SSL Forward Proxy
 - SSL Inbound Inspection
 - Master key storage
 - Private key storage



For a Tech Doc about this topic, log into Live and search for "Device > Setup > HSM"

Module Summary

Now that you have completed this module, you should be able to:

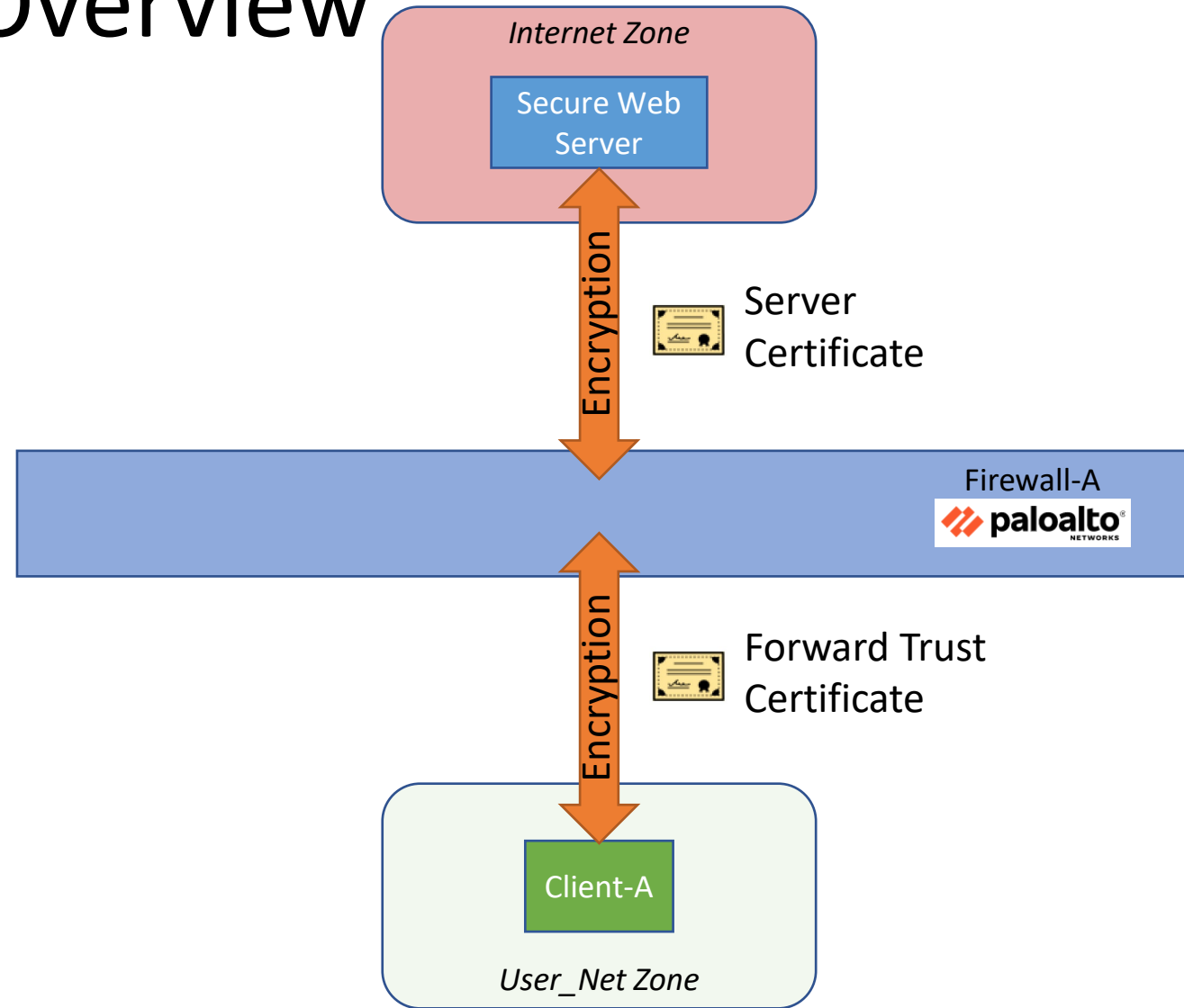
- Review fundamental SSL concepts and operation
- Create and manage certificates using the web interface
- Configure SSL/TLS forward proxy decryption
- Configure SSL/TLS inbound inspection decryption
- Prevent decryption for specific traffic
- View information and troubleshoot SSL/TLS issues using the CLI and logs
- Identify decryption configuration considerations
- Configure SSH decryption
- List other available decryption methods



Questions



Lab 12: Overview



Lab 12: Using Decryption to Block Threats in Encrypted Traffic

- Load a lab configuration
- Test the firewall without decryption
- Create a self-signed certificates for trusted connections
- Create A self-signed certificates for untrusted connections
- Create and test a Decryption policy rule for outbound traffic
- Test outbound Decryption policy rule
- Export the firewall certificate and import to Firefox
- Test outbound Decryption policy again
- Review firewall logs
- Exclude URL categories from decryption using a No-Decrypt rule
- Test the No-Decrypt rule

End Content

Unit 13 Homework

- Always check Blackboard for assignments and when they are due.
- Unit 13 Lab 1: Palo Alto Lab 12: Using Decryption to Block Threats in Encrypted Traffic
- Unit 13 Assignment 1: Github Profile
- Unit 13 Assignment 2: LinkedIn Profile
- Unit 13 Quiz